

Reading Concordances with Algorithms: a training course in key corpus linguistics methodology

Reading with algorithms: FlexiConc

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being in some square, the other exhibitions were IN houses, old shops, small factory spaces, offices and
might be described as 'wet' or 'dry' (W or D) and IN a humid temperate climate such as Britain expect
a powerful message put across by the curriculum IN most schools is that knowledge can be divided into
as like the trouble we had half an hour ago, when IN fact there were three lines, sections of which tran
opportunities for our **READING** arise naturally IN the course of the day; for instance, in the activity
traditional hand-made **CONCORDANCES** IN several ways. One is that they will be much more
about Milan. 'How old were you when you arrived IN Italy?' The American journalist broke into his throu
of Modern Art, keen in discerning what was good IN **THE** arts of many ages and styles. As for writers
who felt it necessary to take aside both captains IN **no** 21ST minute and warn them about each team
region has undergone several media revolutions IN the last **CENTURY**. In principle, now that over
to some extent the ownership of the biggest hits IN each year), the aggregate market share of the top
focus from arts education in general, to one area IN particular, namely English lessons and the fiction
the horse was, shall I say dreamed up, erm and IN fact we had one or two horses, Clydesdale horses
wooded woods. It was merely that he never wanted IN a tournament round to risk anything which might

Orientation

What we have learned so far ...

- Reading concordances with (and without) the help of software tools
- How to identify recurrent patterns and interpret them
- Strategies and algorithms for organising concordance lines

What we're going to look at now ...

- A mathematical framework for reading concordances with algorithms
- What further algorithms could be useful?
- How can we document the concordance reading process?

Our goal

Goal: Develop mathematical framework for computational algorithms used to organise concordance lines in the concordance reading process.

- Tightly connected to kwic-style display of concordance lines
- **Limitation:** only algorithms that can be understood as a rearrangement of concordance lines → makes generic UI possible
- Focus on three main general strategies in current practice

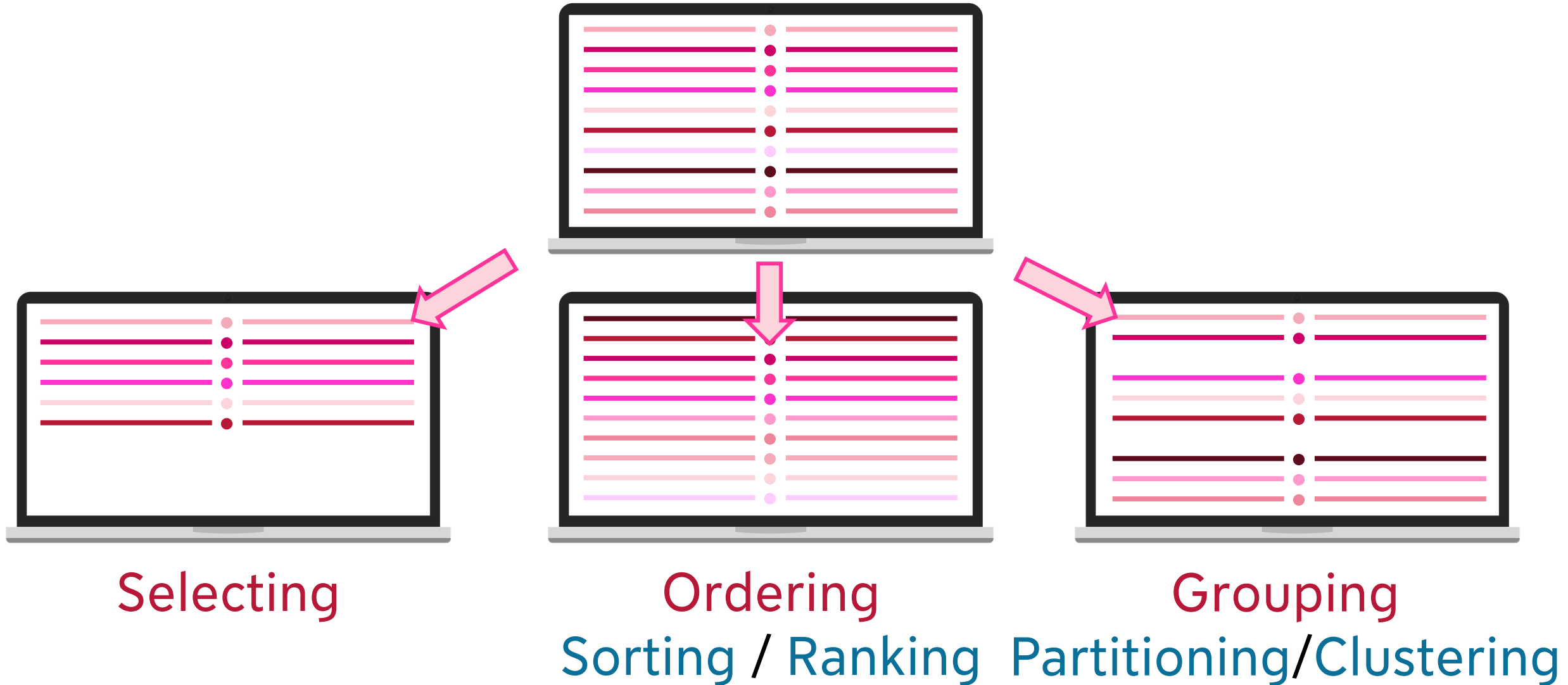
But why?

Why not just invent some really cool new algorithms
and implement them in #FAUbox or EverConc?

But why?

- Community needs general framework, not another app
- Make novel algorithms available in many (all) concordancers
- Specify precise semantics of algorithms
- Combining algorithms: limitations
- Consistent visualisation in UI
- Formal model of organising concordances
- Research documentation

Concordance reading strategies



From strategies to algorithms

1. Selecting

- “zoom in” on a **subset** of the concordance lines
- either focus on specific pattern or discard unwanted lines

2. Ordering

- change **sequential ordering** of concordance lines in the display
- may want to combine multiple ordering criteria to break ties

3. Grouping

- form groups of related concordance lines
- requires **structured kwic display** in UI

From strategies to algorithms

1. Selecting

- “zoom in” on a **subset** of the concordance lines
- either focus on specific pattern or discard unwanted lines

2. Ordering

- a) **Sorting**: reorder lines based on pairwise comparison
- b) **Ranking**: reorder lines based on preference score
- combine multiple Sorting and Ranking algorithms to break ties

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- a) **Partitioning**: groups based on common property
- b) **Clustering**: (nested) groups based on pairwise similarity
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Combining algorithms

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subset

arrangement

Methodology: a reading procedure (Sinclair 2003)

arrangement

Step	Summary
1. Initiate	Decide on the strongest patterns to the left or right of the node (e.g., recurring words, word forms, or parts of speech.)
2. Interpret	Find a hypothesis that links (most of) the repeated words. The link could be lexical, semantic, or syntactic.
3. Consolidate	Look for further evidence to support the hypothesis (e.g., similar meanings expressed in different ways, words in more distant positions.)
4. Report	Write out an explicit, testable version for the future.
5. Recycle	Repeat steps 1–4 with the next most important pattern. Continue until you cannot find any more repeated patterns. Classify the remaining lines as either unusual or a potential pattern that is rare in this set of lines.
6. Result	Make a list of final hypotheses.
7. Repeat	Repeat with a new sample to confirm, extend, or revise your hypotheses.

subset

The analysis tree



Our model of concordance reading: **analysis tree**

- **Selecting** creates branches that “zoom in” on a subset of lines
→ **subset node**
 - investigate different patterns with each branch
 - turns history of operations into a hierarchical tree structure
- **Ordering** and **Grouping** algorithms change how this subset is displayed to the analyst → **arrangement node**
 - different arrangements of the same subset are siblings in the tree
 - default arrangement: original corpus order
- Each node corresponds to a **concordance view**
 - defined as subset of concordance lines + their arrangement

Combining algorithms

- Subset operations can either be nested or form different branches
 - consecutively narrow down to smaller subsets → structure of tree
- Arrangement operations form alternative views of a subset
 - consequence: can't "stack" multiple arrangements
 - combine sequence of **Ordering** algorithms as tie-breakers
 - multiple **Grouping** algorithms cannot (or should not) be combined
- Arrangement node = ≤ 1 **Grouping** + sequence of **Ordering** algo's
 - **Grouping** determines main structure, **Ordering** applies within groups

Corpus

DNOV-CWB

Analysis Tree

Query

"eyes" %c

Partition / Cluster ▼



Make selection ▼

♥ Bookmark

Welcome to
CandyConc

images do not represent actual gameplay

Corpus DNOV-CWB

root node

Analysis Tree

Query result (4031)

Query "eyes" %c

Partition / Cluster ▼

⊕

Make selection ▼

♥ Bookmark

1	BH 1	ing on the gunwales of barges and small boats . Fog in the	eyes	and throats of ancient Greenwich pensioners , wheezing by
2	BH 3	m very cheerful , but I can not quite help their coming to my	eyes	. There ! I have wiped them away now and can go on again
3	BH 3	of the coach , brushed one of his large furry cuffs across my	eyes	(but without hurting me) , and showed me that it was wet .
4	BH 3	e greatest astonishment . But I thought that he had pleasant	eyes	, although he kept on muttering to himself in an angry man
5	BH 3	osegay of geraniums and told me I had been the light of his	eyes	-- indeed the old man said so!-- what a heart I had then ! Ar
6	BH 3	out it than I ought to have been ; and when I had cooled my	eyes	with lavender water , it was time to watch for London . I was
7	BH 3	n a beautiful girl ! With such rich golden hair , such soft blue	eyes	, and such a bright , innocent , trusting face ! " Miss Ada r ,
8	BH 3	ut it ; and the fire , which had left off roaring , winked its red	eyes	at us -- as Richard said -- like a drowsy old Chancery lion .
9	BH 3	fore it was quite said and whispered . His lordship , with his	eyes	upon his papers , listened , nodded twice or thrice , turned c
10	BH 4	Mr. Kenge , standing with his back to the fire and casting his	eyes	over the dusty hearth - rug as if it were Mrs. Jellyby 's biogr
11	BH 4	inutive , plump woman of from forty to fifty , with handsome	eyes	, though they had a curious habit of seeming to look a long
12	BH 4	thing , " Go along , you naughty Peepy ! " and fixed her fine	eyes	on Africa again . However , as she at once proceeded with
13	BH 4	ing him to me ? " Shaking her golden hair , Ada turned her	eyes	upon me with such laughing wonder that I was full of wonde
14	BH 4	died , she remembered how the tears would come into her	eyes	when she spoke of him and of the noble generosity of his c
15	BH 4	was painfully awake and vainly tried to lose myself , with my	eyes	closed , among the scenes of the day . At length , by slow c
16	BH 4	nd day was feebly struggling with the fog when I opened my	eyes	to encounter those of a dirty - faced little spectre fixed upon
17	BH 5	Richard , answering what he saw in Ada 's compassionate	eyes	, took the opportunity of laying some money , softly and unc
18	BH 6	of themselves , would have made a pretty large town in my	eyes	; and at last we got into a real country road again , with win
19	BH 6	ning sudden in his manner and a pleasant expression in his	eyes	images do not represent actual gameplay on
20	BH 6	, glancing at Richard and Ada , who entreated me with their	eyes	to speak] , " that perhaps she was a little unmindful of her h

root node

Analysis Tree

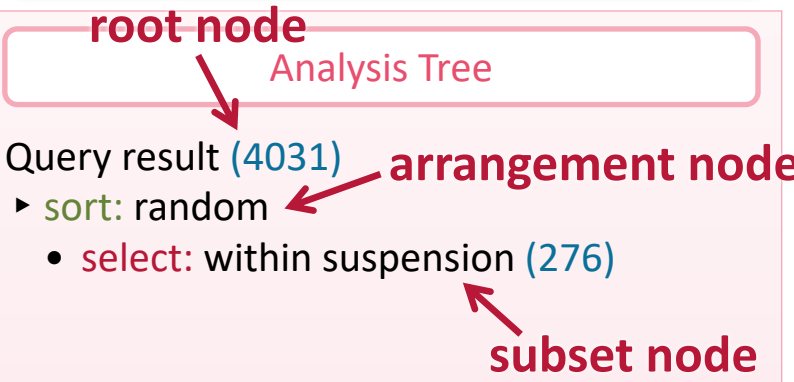
Query result (4031)

• sort: random

arrangement node

151	MC 39	you my word , as if he had been Mark Tapley . I opened my	eyes	, I can tell you , when he did so ; but I could not fancy I had
152	DS 29	ifidence of many years is destroyed in an instant , when my	eyes	are opened all at once , and when I find you revealed in yo
153	BH 55	hitened wall , leans her honest forehead upon it , wipes her	eyes	with her serviceable grey cloak , and quite enjoys herself lik
154	DS 50	lness that lies to lightly there ! Ay , look down on the closed	eyes	, Walter , with a proudly tender gaze ; for in all the wide wid
155	OT 17	npulses of passion or feeling , which , presented before the	eyes	of mere spectators , are at once condemned as outrageous
156	DS 33	g her head after resting it for a moment on both hands , her	eyes	met those of Harriet . In a moment , Harriet was at the door
157	BH 23	aid Mrs. Jellyby , still opening her letters , casting her bright	eyes	smilingly over them , and sorting them as she spoke] , " tha
158	LD 9	ad had no sight for the tears now standing in the colourless	eyes	; though he had had no hearing for the sob that checked the
159	TTC 21	man spoke when they entered the wine - shop , though the	eyes	of every man there were turned upon them . " Good day , g
160	OMF 22	ere , he now directed a scowling look into Fledgeby 's small	eyes	for the effect of the opposite treatment . Satisfied by what h
161	OT 14	s home four times a - year , that it brought the tears into her	eyes	to talk about them . When the old lady had expatiated , a lo
162	HT 13	s Rachael . Seen across the dim candle with his moistened	eyes	, she looked as if she had a glory shining round her head . I
163	OMF 16	Old Betty was proud , and said it with a sparkle in her bright	eyes	. ' Yes , but there are some little comforts that you would n't
164	DS 51	t Mr Pitt , upon the bookcase , represents it . That there are	eyes	in its own map , hanging on the wall . ' An unusually cold sp
165	BH 28	e hand with which she has caught it , and standing with her	eyes	fixed on the fire , puts it about and about between her own
166	PP 54	here was enough of the cannibal in the young gentleman 's	eyes	to render the compliment a double one . ' Dear me , Joseph
167	GE 17	ening that she had curiously thoughtful and attentive eyes ;	eyes	that were very pretty and very good . It came of my lifting up
168	LD 5	d to you to suspect-- ' At the word Suspect , she turned her	eyes	momentarily upon her son , with a dark frown . She then su
169	DC 16	ting themselves - that they seemed to twinkle instead of his	eyes	quite
170	DS 10	d a most excellent man , Sir [, ' pursued Walter , raising his	eyes	with a look of entreaty in the Captain 's behalf] , " was so gc

1	LD 16	stopping for a moment with a good - humoured laugh in his	eyes	, and laying his closed right hand , with its peculiar suppleness
2	BR 37	, I have slept so soundly [, ' said Lord George , rubbing his	eyes	and looking round the room] , ' that I do n't remember quite
3	DS 47	when you give Mrs Dombey to understand [' --the light in his	eyes	fell upon her as he separated his words each from each , w
4	DS 61	sh sorry [, ' said Cousin Feenix , lifting his wristbands to his	eyes	in the simplest manner possible , and without the least conc
5	OT 37	ppose , a married man [, ' replied Mr. Bumble , shading his	eyes	with his hand , and surveying the stranger , from head to fo
6	NN 22	e place . ' ' Well , when I was there [, ' resumed Smike ; his	eyes	sparkling at the prospect of displaying his abilities] ; ' I coul
7	DS 40	sh a nature now , ' --he [fancied he saw tears glistening in her	eyes	, and he thought , complacently , that he had forced them fr
8	BH 4	Mr. Kenge , standing with his back to the fire and casting his	eyes	over the dusty hearth - rug as if it were Mrs. Jellyby 's biogr
9	DS 17	oked pleasant . ' You see [, ' began the Captain , rolling his	eyes	slowly round the little room , and taking in as much of it as I
10	OMF 3	Eugene , leaning back , folding his arms , smoking with his	eyes	shut , and speaking slightly through his nose] , ' of Energy .
11	DC 39	I do n't think , Trotwood [, ' returned Agnes , raising her soft	eyes	to mine] , ' I would consider that . Perhaps it would be bette
12	BH 9	ray and regarding me , as I again strangely felt , though my	eyes	were not directed to him , with his late intent look] , " I am a
13	BH 23	aid Mrs. Jellyby , still opening her letters , casting her bright	eyes	smilingly over them , and sorting them as she spoke] , " tha
14	DS 10	d a most excellent man , Sir [, ' pursued Walter , raising his	eyes	with a look of entreaty in the Captain 's behalf] , ' was so ge
15	DS 20	r . Notwithstanding which , Ma'am [, ' he added , raising his	eyes	from his plate , and casting them across Princess 's Place ,
16	BH 26	wo n't sit down like a Christian [, " says Mr. George with his	eyes	musingly fixed on Judy] , " I ca n't comprehend . " " She kee
17	OMF 38	ollecting [, ' said Mrs Boffin , with her work abandoned , her	eyes	upon the fire , and her chin upon her hand] , ' do you recoll
18	DS 22	onder [, ' said Biler , blubbering outright , and smearing his	eyes	with his coat - cuff] , ' that I have n't been and drowned my
19	OMF 25	into tears . ' I think , R.W. [, ' cried Mrs Wilfer , lifting up her	eyes	it wo
20	DS 46	uttered the old woman , watching her daughter with her red	eyes	; ' so easy and so trim a - horseback , while we are in the m



root node

Analysis Tree

Query result (4031) arrangement node

▶ sort: random

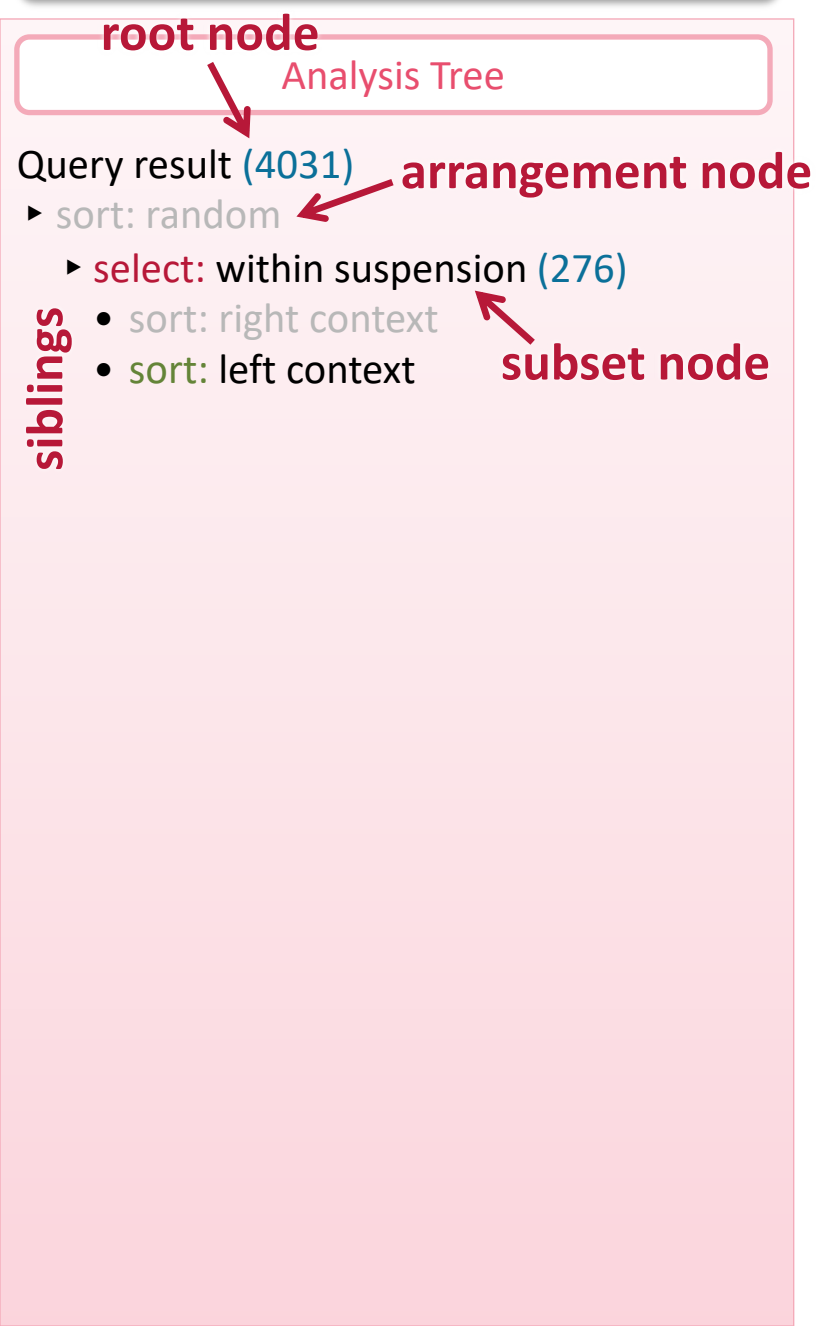
▶ select: within suspension (276)

• sort: right context

subset node

101	LD 14	le Dorrit began , sitting before the pale fire , and raising her	eyes	again to the face which in its harmonious look of interest ,
102	NN 31	. ' Oh [! ' said Newman , throwing some intelligence into his	eyes	all at once , and dropping them on his master] , ' I thought
103	BH 47	orge [, " says Allan , following the direction of the trooper 's	eyes	along the entry] , " have not been much acquainted , as yo
104	OMF 25	into tears . ' I think , R.W. [, ' cried Mrs Wilfer , lifting up her	eyes	and apostrophising the air] , ' that if you were present , it w
105	TTC 44	gement that may be [, " said Miss Pross , striving to dry her	eyes	and compose herself] , " I have no doubt it is best that Mrs.
106	GE 39	ng , Pip [, " said he , after again drawing his sleeve over his	eyes	and forehead , as the click came in his throat which I well r
107	BH 14	, my dear . I suppose [, " said Mr. Turveydrop , shutting his	eyes	and lifting up his shoulders with modest consciousness] , "
108	HT 23	Yonder , among the trees [, ' he added , having lifted up his	eyes	and looked about ; for he had watched her closely until nov
109	BR 37	, I have slept so soundly [, ' said Lord George , rubbing his	eyes	and looking round the room] , ' that I do n't remember quite
110	OCS 6	the first is dead , sweet Nell [, ' said Quilp , wrinkling up his	eyes	and luring her towards him with his bent forefinger] , ' to be
111	OMF 61	ching her cheek , with a grave smile , as she cast down her	eyes	and pouted again] ; ' look at me . I want to speak to you . ' "
112	MC 9	dergone in consequence [, ' cried Mrs Todgers , raising her	eyes	and shaking her head] , ' no one would believe ! ' ' Just like
113	BH 15	go out ? " " As often as I can [, " said Charley , opening her	eyes	and smiling] , " because of earning sixpences and shillings
114	LD 70	e best , my own [, ' she answered , laughing with glistening	eyes	and standing on tiptoe to kiss him] , ' if you will only humou
115	DS 31	e stooping of his neck , which the revelations making by his	eyes	and teeth proclaim to be a lie] , ' I am proud to see that my
116	MC 50	e time now drawing on [, ' said the old man , hurriedly -- his	eyes	and whole face brightening as he spoke--'will] make amenc
117	DS 20	ng his head at the unconscious Miss Tox , while his starting	eyes	appeared to make a leap towards her] , ' you might do that
118	BH 58	st I could do . " " You find me [, " says Sir Leicester , whose	eyes	are much attracted towards him] , " far from well , George F
119	BH 41	he returns without looking up from the ground on which her	eyes	
120	LD 48	sence of mind . ' Pray [, ' asked Lord Decimus , casting his	eyes	

101	OMF	59	es n't know anything about it [, ' with a sob , and wiping her	eyes	[, ' why , it wo n't matter to him . I see the service in the Pray
102	DC	60	' of doing good ; and if I could spare my brother [, ' with her	eyes	upon me] , ' perhaps the time could not . ' ' What I am , you
103	DS	27	looking round . ' Shall I tell you [, ' she continued , with her	eyes	fixed on her mother] , ' who already knows us thoroughly , a
104	OMF	58	a wink . ' I must own [, ' returned the dressmaker , with her	eyes	upon her work] , ' that we are not good friends at present . '
105	DC	45	' said Mrs. Markleham , mechanically following her with her	eyes	[, ' to find a man at Doctor Strong 's time of life , with the stre
106	DS	52	lth [, ' said the Grinder , glancing at Alice , who sat with her	eyes	fixed , as it seemed to him , on the wall behind him , but in i
107	OMF	15	at then ? ' ' Why then [, ' said Mrs Boffin , speaking with her	eyes	closed , and her left hand thoughtfully touching her brow] ,
108	LD	14	g at him in a suppliant way , and gradually withdrawing her	eyes	as he looked at her] , ' whether you would advise me what
109	DS	23	brig as a lad is capable on . Wal'r [, ' said the Captain , his	eyes	glistening with the praise of his young friend , and his hook
110	PP	23	great advantage of this , Mr. Weller [, ' continued Job , his	eyes	filling with tears as he spoke] , ' will be , that I shall be able
111	PP	27	ose . ' Certainly , Sam , certainly [, ' said Mr. Pickwick , his	eyes	glistening with delight at this manifestation of filial feeling or
112	MC	50	a time now drawing on [, ' said the old man , hurriedly -- his	eyes	and whole face brightening as he spoke--'will] make amenc
113	MC	19	o limitation , there is positively NO limitation [' --opening his	eyes	wide , and standing on tiptoe--'in] point of expense ! I have
114	NN	22	e place . ' ' Well , when I was there [, ' resumed Smike ; his	eyes	sparkling at the prospect of displaying his abilities] ; ' I coul
115	OCS	3	said the dwarf , who with his hand stretched out above his	eyes	had been surveying the young man attentively] , ' that shou
116	OMF	42	d the Reverend Frank Milvey , brushing his hand across his	eyes	[, ' when it has that homely figure on it . Richer , I think , tha
117	BR	76	ith , stopping in a narrow count to pass his hand across his	eyes	[, ' I could better afford to lose than Barnaby . We have alwa
118	BR	36	he secretary , drawing his sleeve in a hasty way across his	eyes	[; ' and why should I regret your having heard them ? ' ' Gas
119	NN	52	ink and cough very much , for he was no smoker) : and his	eyes	
120	OMF	39	oodly silence , with his hands clasped behind him , and his	eyes	



No.	Query result	No. of occurrences	Percent
1	PRP\$	211	76.45%
2	JJ	37	13.41%
3	VBG	9	3.26%
4	NN	5	1.81%
5	DT	4	1.45%
6	WP\$	3	1.09%
7	IN	2	0.72%
8	POS	2	0.72%
9	VBN	2	0.72%
10	CC	1	0.36%

root node

Analysis Tree

arrangement node

subset node

siblings

Query result (4031)

▸ sort: random

▸ select: within suspension (276)

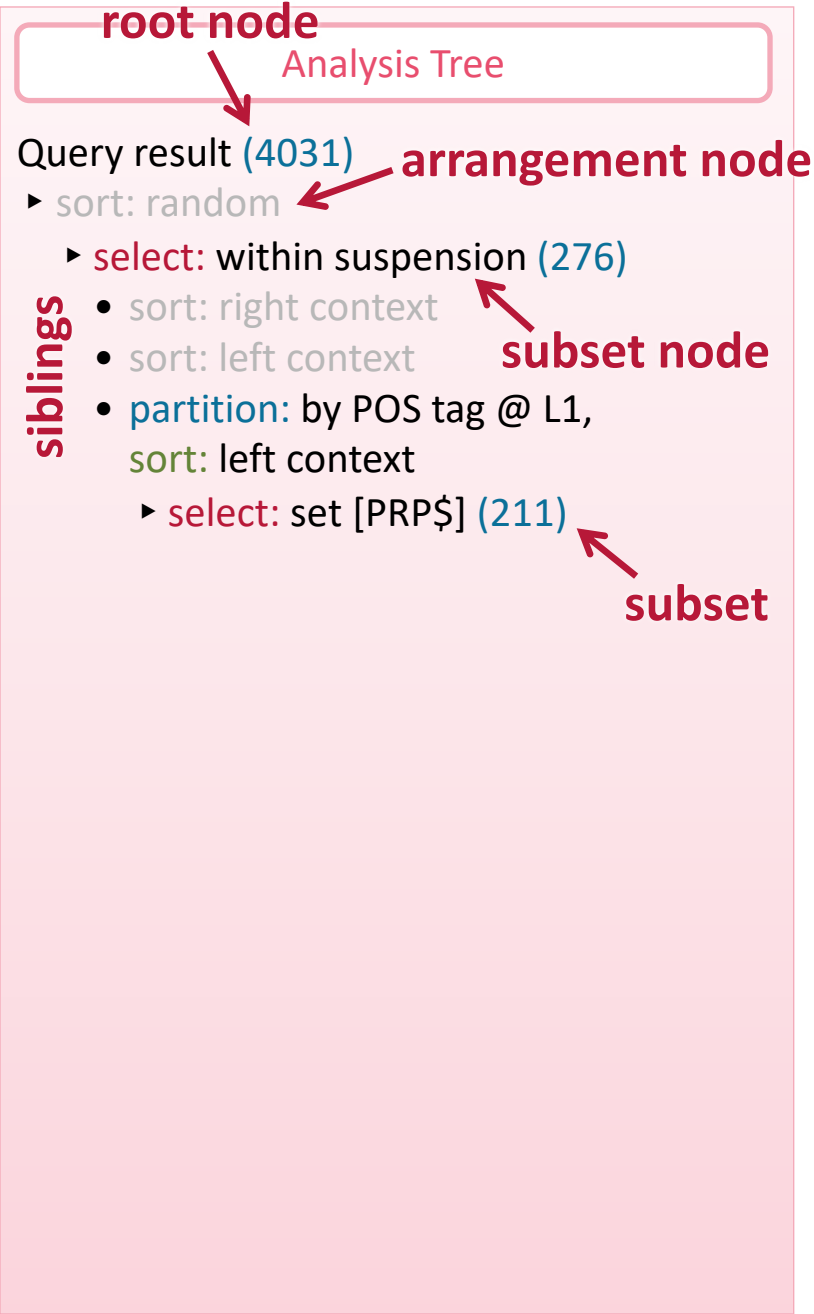
• sort: right context

• sort: left context

• partition: by POS tag @ L1,

sort: left context

1	OMF 38	collecting [, ' said Mrs Boffin , with her work abandoned , her	eyes	upon the fire , and her chin upon her hand] , ' do you recoll
2	BH 49	red thousand thanks , old fellow [, " says Mrs. Bagnet , her	eyes	sparkling with pleasure and her hand stretched out to him]
3	BH 22	brought into my head , master [, " returns the woman , her	eyes	filling with tears] , " when I look down at the child lying so .
4	OT 24	' she cried , jumping fiercely up : her face flushed , and her	eyes	starting from her head--'I] robbed her , so I did ! She was n
5	OMF 42	netimes I have fancied , sir [, ' said Bella , turning away her	eyes]	, ' that your great regard for Mrs Boffin is a very powerful m
6	DS 18	l , if ever there was one [, ' said Mrs Chick , drying both her	eyes	at once with great care on opposite corners of Miss Tox 's h
7	OT 7	deed . ' Ah [!' said Mr. Bumble , when the lady brought her	eyes	down to earth again] ; ' the only thing that can be done now
8	LD 2	' ' Not [, ' said Pet , apologetically and embarrassed by her	eyes]	, ' not , of course , that we are any company to you , or that
9	BH 8	ng that her spectacles were made the less engaging by her	eyes	being what Ada called] " choking eyes " , " meaning very pr
10	BH 23	Miss Summerson [, " said Mrs. Jellyby , sweetly casting her	eyes	for a moment on me and considering where to put the parti
11	PP 22	tared wildly at her . ' Wretch [, ' said the lady , covering her	eyes	with her hands] , ' what do you want here ? ' ' Nothing , ma'
12	HT 9	not yours . ' ' Dear Miss Louisa [, ' said Sissy , covering her	eyes	, and sobbing yet] ; ' I came home from the school that afte
13	OMF 61	ching her cheek , with a grave smile , as she cast down her	eyes	and pouted again] ; ' look at me . I want to speak to you . ' '
14	LD 55	eneral . ' ' Mr Dorrit [, ' said Mrs General , casting down her	eyes	as she rose with a bend] , ' must ever claim my homage an
15	DS 58	. ' I need not remind you [, ' said Harriet , casting down her	eyes	upon her black dress] , ' through what means our circumsta
16	OMF 25	ith stately resignation as she shook her head and dried her	eyes]	; ' we have at present no stipendiary girl to do so . We have
17	ED 13	ue . ' If we knew yesterday [, ' said Rosa , as she dried her	eyes]	, ' and we did know yesterday , and on many , many yester
18	DS 2	always be a balm and comfort to me [, ' here she dried her	eyes]	; ' it becomes necessary to provide a Nurse . ' ' Oh ! Ah ! ' s
19	TTC 44	gement that may be [, " said Miss Pross , striving to dry her	eyes	
20	DC 51	a gleam of light upon me , Trot [, ' said my aunt , drying her	eyes]	



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17	ED 13	ue . ' If we knew yesterday [, ' said Rosa , as she dried her	eyes]	, ' and we did know yesterday , and on many , many yester
18	DS 2	always be a balm and comfort to me [, ' here she dried her	eyes]	; ' it becomes necessary to provide a Nurse . ' ' Oh ! Ah ! ' s
19	TTC 44	gement that may be [, " said Miss Pross , striving to dry her	eyes	
20	DC 51	a gleam of light upon me , Trot [, ' said my aunt , drying her	eyes]	

siblings

root node

Analysis Tree

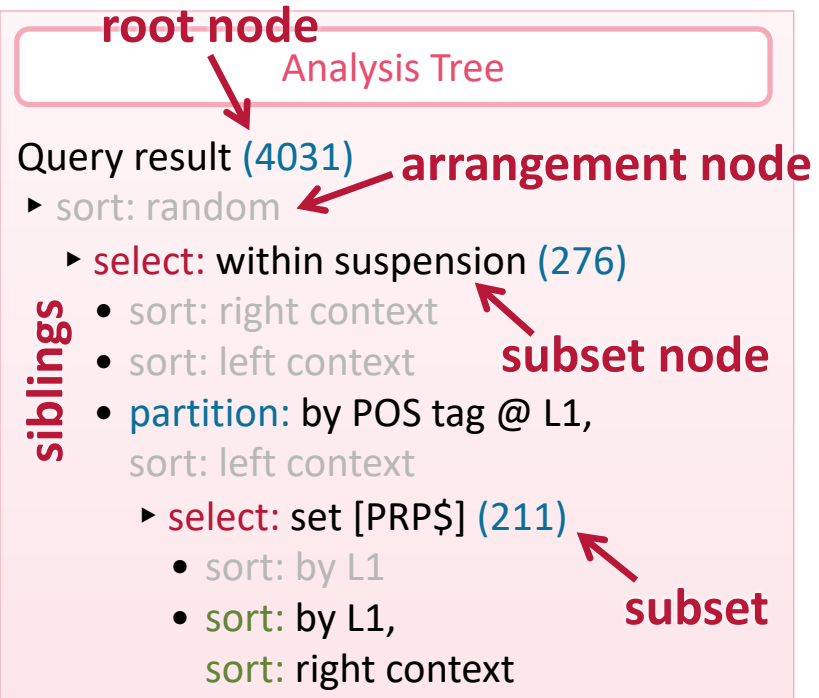
Query result (4031) arrangement node

- ▶ sort: random
- ▶ select: within suspension (276)
- sort: right context
- sort: left context
- partition: by POS tag @ L1, sort: left context
 - ▶ select: set [PRP\$] (211)
 - sort: by L1

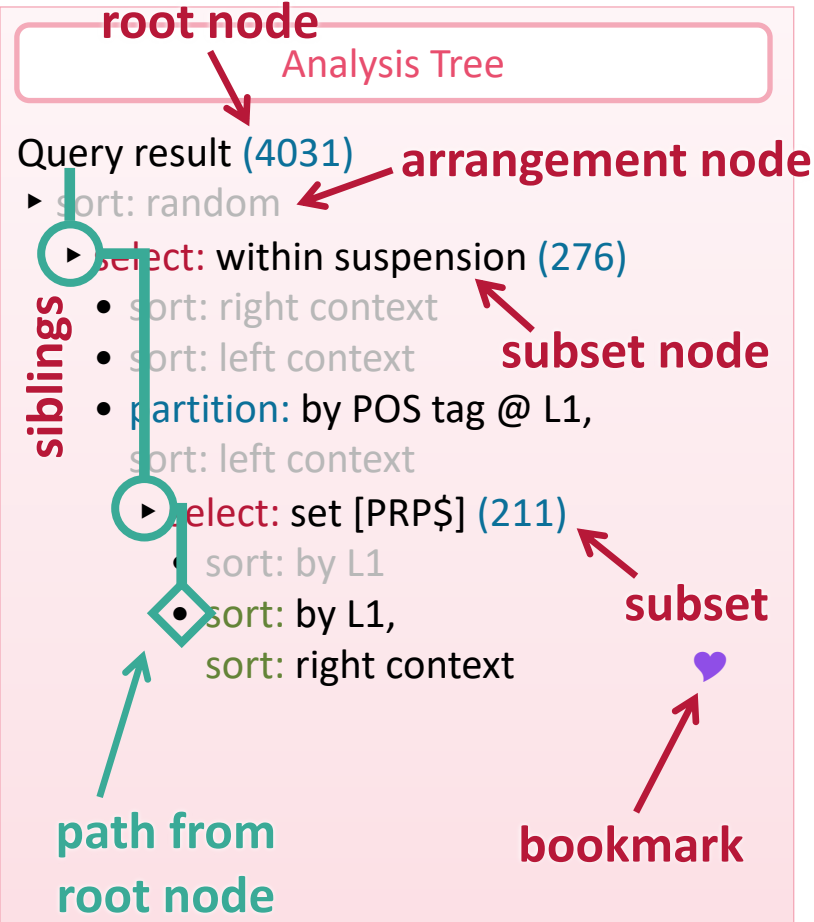
subset node

subset

1	BH 30	ith us . " Well , my dear Esther [, " she returned , drying her	eyes	, " I must do the best I can and trust to my dear Prince neve
2	HT 4	circus [, ' muttered Louisa , haughtily , without lifting up her	eyes	, ' and father caught us . ' ' And , Mrs. Gradgrind [, ' said he
3	ED 13	rue . ' If we knew yesterday [, ' said Rosa , as she dried her	eyes	, ' and we did know yesterday , and on many , many yester
4	NN 49	gives me great pain [, ' continued Mrs Nickleby , drying her	eyes	, ' great pain ; but do n't hurt a hair of his head , I beg . On r
5	DS 3	my Mama [, ' returned the child , with tears springing to her	eyes	, ' in any frock . ' ' But people put on black , to remember pe
6	NN 34	culty . And even this [, ' said Madame Mantalini , wiping her	eyes	, ' is not the worst . He took some papers of value out of my
7	NN 37	her . I think I see him now [! ' said Mrs Nickleby , wiping her	eyes	, ' looking at me while I was talking to him about his affairs ,
8	LD 2	' ' Not [, ' said Pet , apologetically and embarrassed by her	eyes	, ' not , of course , that we are any company to you , or that
9	HT 32	this . ' " Deed , I am loath [, ' returned Rachael , drying her	eyes	, ' that any here should see me like this ; but I wo n't be see
10	LD 41	ha -- on this or any other occasion [, ' Mrs General shut her	eyes	, ' that I -- ha hum -- am not pleased with you . You make M
11	BR 51	ately , Varden [, ' said his wife , with her handkerchief to her	eyes	, ' that in case any more disturbances should happen -- whi
12	OMF 28	n vain to sustain it ; for the look dropped as it lighted on her	eyes	, ' that it might be so superfluous as to be almost impertiner
13	OMF 42	netimes I have fancied , sir [, ' said Bella , turning away her	eyes	, ' that your great regard for Mrs Boffin is a very powerful m
14	DC 48	gently trying to remove the handkerchief she pressed to her	eyes	, ' this is not only very ridiculous of you , but very wrong . In
15	DC 45	' said Mrs. Markleham , mechanically following her with her	eyes	, ' to find a man at Doctor Strong 's time of life , with the stre
16	MC 40	d kindly . ' Which shows [, ' said Mrs Gamp , casting up her	eyes	, ' what a little way you 've travelled into this wale of life , my
17	OCS 41	d . ' Oh Kit [! ' said his mother , with her handkerchief to her	eyes	, ' what have you done ! I never can go there again -- never
18	LD 14	th all the earnestness of her soul looking steadily out of her	eyes	, ' what I am going to ask you not to do ? ' ' I think I can .
19	DC 51	a gleam of light upon me , Trot [, ' said my aunt , drying her	eyes	
20	OMF 59	es n't know anything about it [, ' with a sob , and wiping her	eyes	



1	BH 30	ith us . " Well , my dear Esther [, " she returned , drying her	eyes	, " I must do the best I can and trust to my dear Prince neve
2	HT 4	circus [, ' muttered Louisa , haughtily , without lifting up her	eyes	, ' and father caught us . ' ' And , Mrs. Gradgrind [, ' said he
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4	NN 49	gives me great pain [, ' continued Mrs Nickleby , drying her	eyes	, ' great pain ; but do n't hurt a hair of his head , I beg . On r
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9	HT 32	this . ' " Deed , I am loath [, ' returned Rachael , drying her	eyes	, ' that any here should see me like this ; but I wo n't be see
10	LD 41	ha -- on this or any other occasion [, ' Mrs General shut her	eyes	, ' that I -- ha hum -- am not pleased with you . You make M
11	BR 51	ately , Varden [, ' said his wife , with her handkerchief to her	eyes	, ' that in case any more disturbances should happen -- whi
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18	LD 14	th all the earnestness of her soul looking steadily out of her	eyes	, ' what I am going to ask you not to do ? ' ' I think I can .
19	DC 51	a gleam of light upon me , Trot [, ' said my aunt , drying her	eyes	our
20	OMF 59	es n't know anything about it [, ' with a sob , and wiping her	eyes	, ' why , it wo n't matter to him . I see the service in the Pray



images do not represent actual gameplay

A mathematical model of organising conc.

- Concordance reading = **analysis tree** consisting of **subset** nodes and **arrangement** nodes
 - subset nodes form the structure of the tree → focus on part of conc.
 - arrangement nodes = alternative views of the same subset (→ parent)
 - arrangement nodes are siblings, subset nodes can be children of either kind
 - root node = subset node corresponding to full concordance
- **Subset** node = Selecting algorithm with parameters
- **Arrangement** node = ≤ 1 Grouping algorithm with parameters + sequence of Ordering algorithms with parameters
 - each arrangement node completely overwrites any prior arrangement

Concordance view

- Each node in the analysis tree represents a **concordance view**, i.e. an arrangement of a subset of the concordance
- Identifies **subset** + **ordering** of conc. lines via **line N°** as unique ID
- With Grouping algorithm: list of **sets** of conc lines specified by line N°
 - **Partitioning**: ordering of the sets ($\neq$ ordering of lines)
 - **Clustering**: ordered tree of the sets + prototypical examples for each set
- Optional: supplementary information on concordance lines, sets, and relevant token spans returned by algorithms

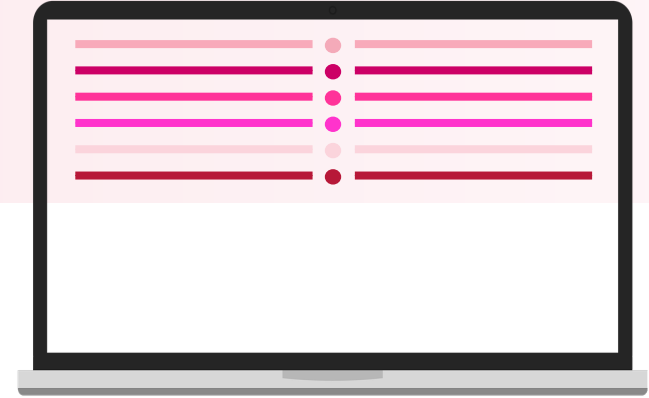
Formalising algorithms

Application of an algorithm is conceptualised mathematically as annotation of concordance lines:

- Selecting = annotation of a **Boolean variable**
- Sorting = annotation of an **integer variable** (surrogate keys)
- Ranking = annotation of a **floating-point variable** (preference scores)
- Partitioning = annotation of a **categorical variable** (set membership)
- Clustering = annotation of **ordered tree** of subsets

→ **concordance view** derived from annotations

Strategy 1: Selecting



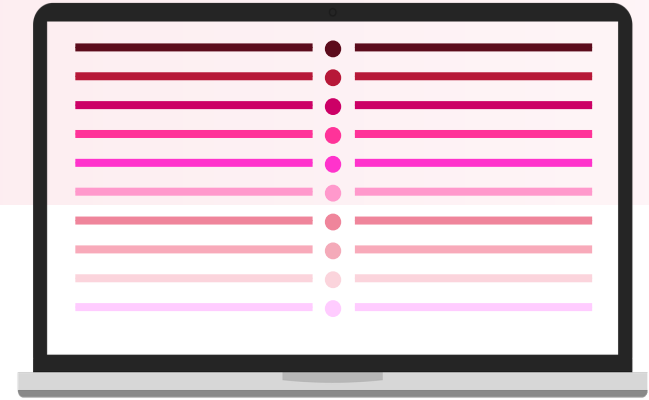
Selecting = add **Boolean variable** to lines

- keep concordance line if value is True
- exclude selected lines by logical negation of variable (all algorithms)

Examples

- selection based on metadata categories or numerical ranges
- select random subset (reproducible, stratified, ...)
- KWIC filter (lines containing certain words, lemmas, POS tags, ...)
- manual selection of individual concordance lines (will get back to this)

Strategy 2: Sorting [Ordering]



Sorting = add **integer variable** with sort keys

- pairwise comparison of lines ($A < B$, $A = B$, $A > B$)
- resulting sort order $\rightarrow$ integers as **surrogate keys** ($A < B$, $A = B$, $A > B$)
- ties ($A = B$) can be broken by additional **Sorting** or **Ranking** algorithms

Examples

- sort alphabetically by left or right context
- sort by POS tag of first (L1) or second (L2) token before node
- sort by numerical (e.g. year) or categorical (e.g. author) metadata
- random shuffling (reproducible)

Strategy 3: Ranking [Ordering]



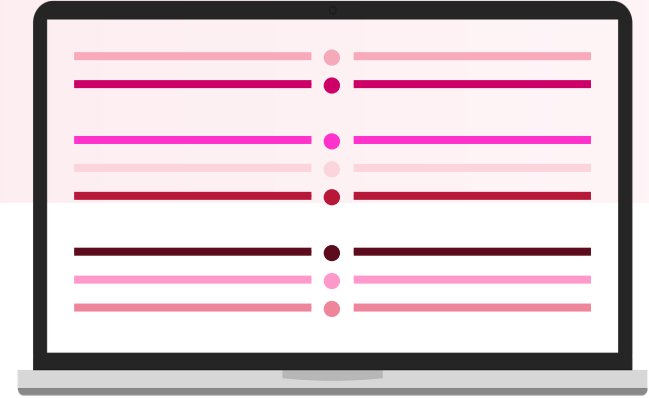
Ranking = add **floating-point variable** to lines

- values = preference score assigned to each line
- decreasing order: lines with highest scores shown at top
- ties can occur e.g. if scores happen to be integer values

Examples

- rank by readability, GDEX algorithm, or fit to selected CEFR level
- rank by frequency of the node string
- rank by similarity to query (for fuzzy search)
- rank by number of significant collocates of the node in context, or by number of user-specified keywords (KWIC grouper)

Strategy 4: Partitioning [Grouping]



Partitioning = add **partition N°** to each line

- each line assigned to exactly one partition, based on a property of the concordance line (not on comparisons)
- also provide ordered list of partitions with display labels, etc.

Examples

- frequency breakdown of node strings
- partition by POS tag of first (L1) or second (L2) token before node
- partition by categorical metadata (variety, genre, speaker info, ...)
- partition by lemma trigram R1+R2+R3 (similar to KWIC patterns)

Strategy 5: Clustering [Grouping]



Clustering = add **ordered tree** over lines

- based on pairwise similarities of concordance lines
- hierarchical clustering → deep tree, flat clustering → single-level tree
- tree describes both nesting and preference order of clusters

Examples

- clustering based on lexical similarity (shared words), semantic similarity (sentence/word embeddings), or syntactic similarity
- clustering based on multiple metadata features → similarity score
- all clusterings: either flat or hierarchical, with different algorithms
- context tree (on POS tags, lemmas, semantic tags, ...)

Aside: Context trees & KWIC patterns

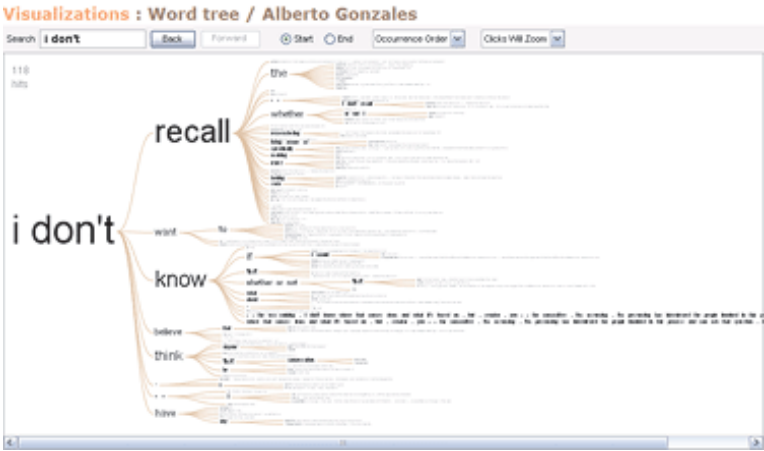


Fig 5: Alberto Gonzales' testimony in 2007.

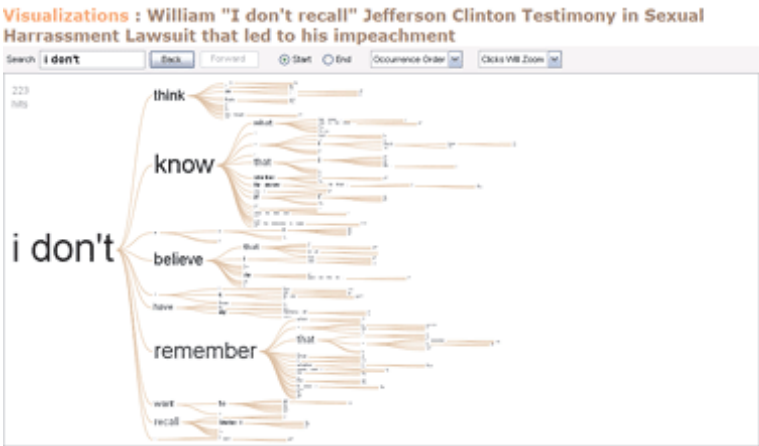
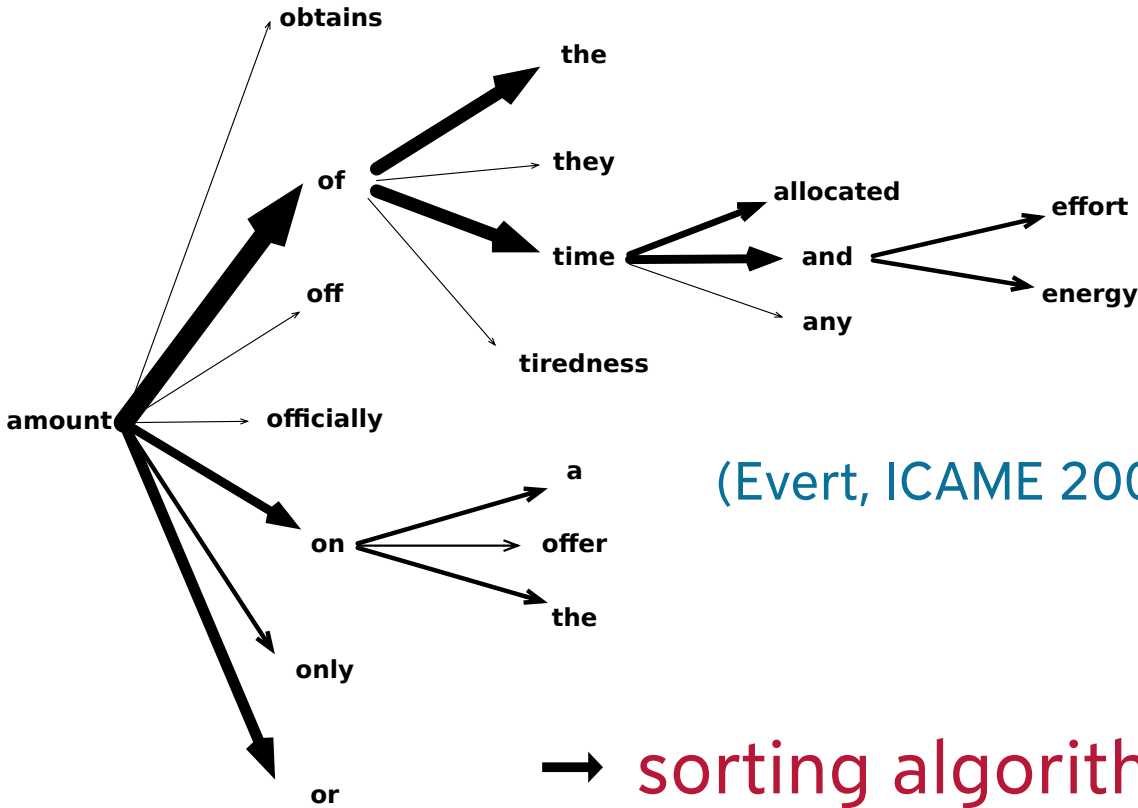


Fig 6: Bill Clinton's testimony in 1998.

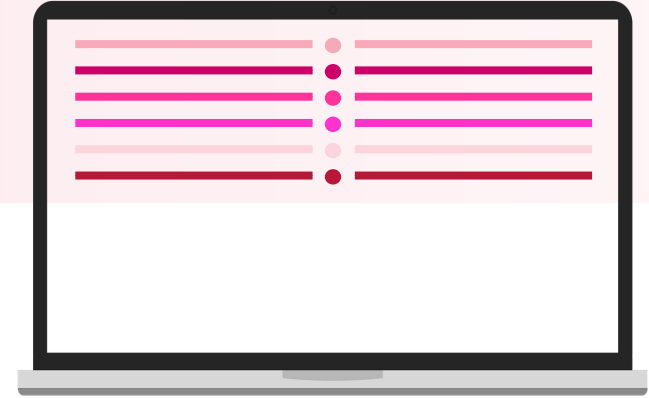


(Evert, ICAME 2007)

The Word Tree
(Wattenberg & Viégas, 2008)

→ sorting algorithm for
concordance reading
(Anthony 2022)
... but 3 different versions

Strategy 1': Manual Selecting



Selecting based on manual choice made by user

- also adds Boolean variable & can be inverted
- direct selection of lines vs. indirect (via set ID or supplementary information)

Examples

- manual validation of concordance lines (e.g. false positives)
- select sets by ID (after Grouping) or range of lines (after Ordering)
- threshold based on ranking scores (returned as supplementary info)
- “interactive” algorithms (e.g. click individual lines or sets) vs. “non-interactive” algorithms (e.g. top- n lines in ranking)

Research documentation

- Analysis tree as **research documentation**
 - concordance view uniquely determined by path from root node
 - bookmarked nodes = basis for interpretation & discussion
- Exact **reproduction** with identical concordance data
 - readers can reproduce published results completely
 - but also explore further concordance views
- **Save / load** analysis tree → continue concordance reading later
- Analysis tree as **template** for concordance organisation
 - “replay” tree on a new concordance (possibly from different corpus)
 - interactive manual selection has to be omitted (→ not meaningful)

Time for a break ...

Coffee



Hands-on: analysis tree in CLiC



UNIVERSITY OF
BIRMINGHAM



Arts & Humanities
Research Council



University of
Nottingham
UK | CHINA | MALAYSIA

<https://clic-fiction.com/>



19C - 19th Century Reference Corpus
29 books,
4,512,568 total words

Load | Merge | Save | Clear | Help

Concordance

FlexiConc

Subsets

Keywords

Counts

clic-analysis-tree_head_19c.json

Welcome to CLiC. The CLiC web app has been developed as part of the [CLiC Dickens project](#), which demonstrates through corpus stylistics how computer-assisted methods

Exercise: body language patterns

- Search words: *cheeks, cheek, neck, fingers, ear*
 - corpus: 19C fiction – focus on non-quotes
- Your task: work through handout & explore FlexiConc algorithms

Showing 1 to 50 of 1,575 entries,
f25a738

ID	Left	Node	Right	Book	In bk.
0	one tinge of crimson flushed the waxen whiteness of her	cheeks;	not one shadow of brown redeemed the pale insipidity of	LadyAud	
1	at half-past nine o'clock, singing a little Scotch melody, her	cheeks	tinged with as delicate a pink as the pale hue	LadyAud	
2	Audley's face; the pretty, roseate flush faded out from her	cheeks,	and left them waxen white, and angry flashes lightened in	LadyAud	
3	were of that sunless, flaxen hue which generally accompanies pink	cheeks	and white eyelashes. She had been a rustic beauty, perhaps	LadyAud	
4	clinging about him, kissed the dirty tears from his faded	cheeks.	¶ "Don't be sorry for me, gran'pa," he said; "I am	LadyAud	
5	The unnatural color still burnt like a flame in her	cheeks;	the unnatural light still glittered in her eyes. The excitement	LadyAud	
6	better nor half the folks as are growned up." ¶ Maggie's	cheeks	began to flush with triumphant excitement. She thought Mr. Riley	mill	
7	lavishing so many warm kisses on it that the waxen	cheeks	had a wasted, unhealthy appearance. ¶ "Did you ever hear the	mill	
8	stared with the roundest of eyes above the reddest of	cheeks;	but was now entirely defaced by a long career of	mill	
9	as much alike as goslings,--a lad with light-brown hair,	cheeks	of cream and roses, full lips, indeterminate nose and eyebrows	mill	
10	for company, and they ate together and rubbed each other's	cheeks	and brows and noses together, while they ate, with a	mill	
11	hands, hurting them with her large rings, and kissing their	cheeks	much against their desire. "Look up, Tom, look up. Boys	mill	
12	said Maggie, giving a little stamp with her foot. Her	cheeks	were quite flushed. ¶ The black locks were so thick, nothing	mill	

FlexiConc

Search in CLiC

Search the corpora:

19C - 19th Century Reference Corpus ✕

Only in subsets:

Non-quotes ▾

Search for terms:

cheeks cheek neck fingers ear

☐ Whole phrase ☒ Any word

Add annotation... ▾

✱ 1 +

Add algorithm... ▾

References

- Anthony, L. (2022). What can corpus software do? In O’Keeffe, A. and McCarthy, M. J., editors, *The Routledge Handbook of Corpus Linguistics*, chapter 9. Routledge, London, 2nd edition
- Evert, S. (2007). Context graphs. Presentation at the *ICAME 2007 Conference*, Stratford upon Avon, UK.
- Wattenberg, M. & Viégas, F. B. (2008). The word tree, an interactive visual concordance. *IEEE Transactions on Visualization and Computer Graphics*, 14(6):1221–1228.